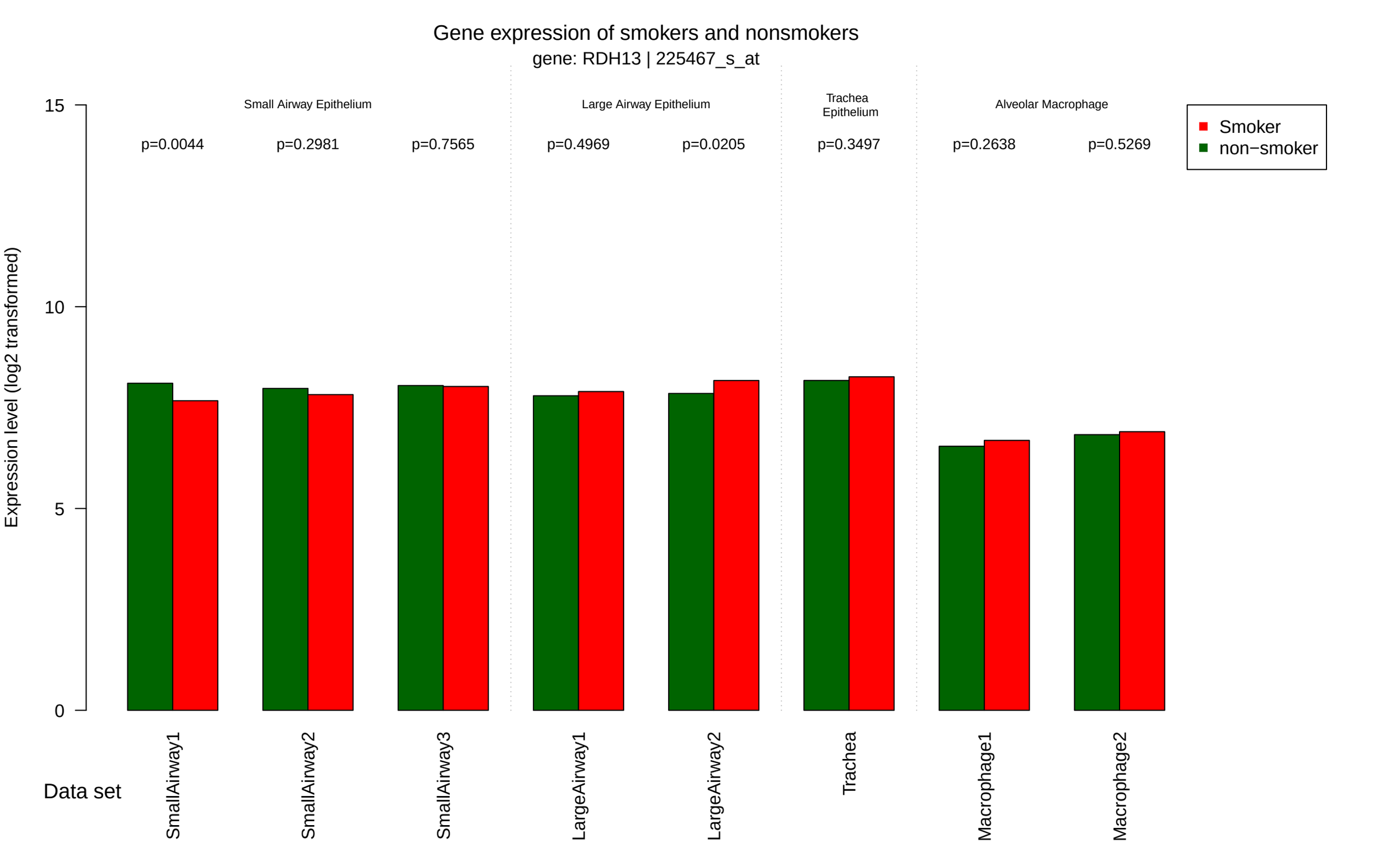
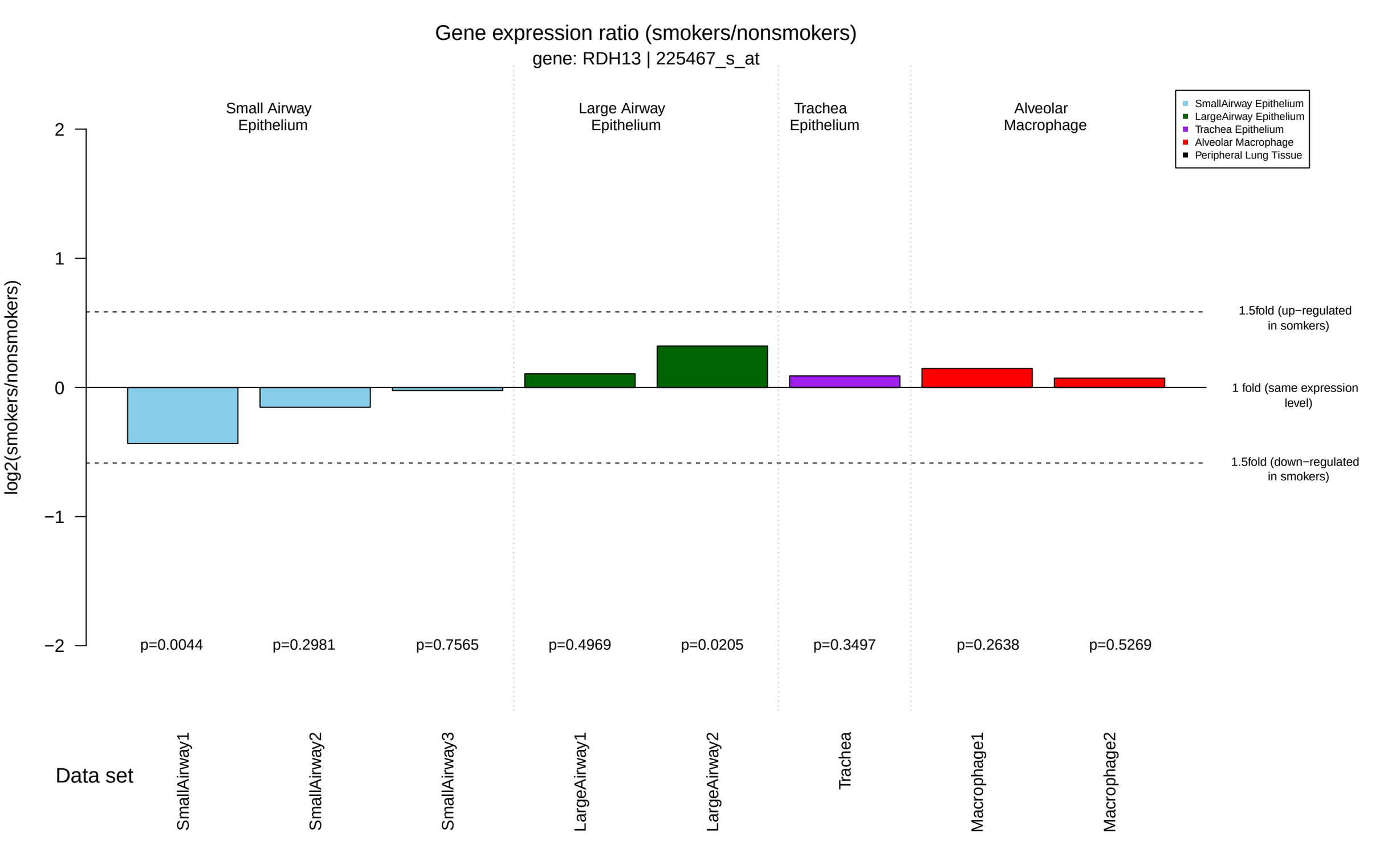
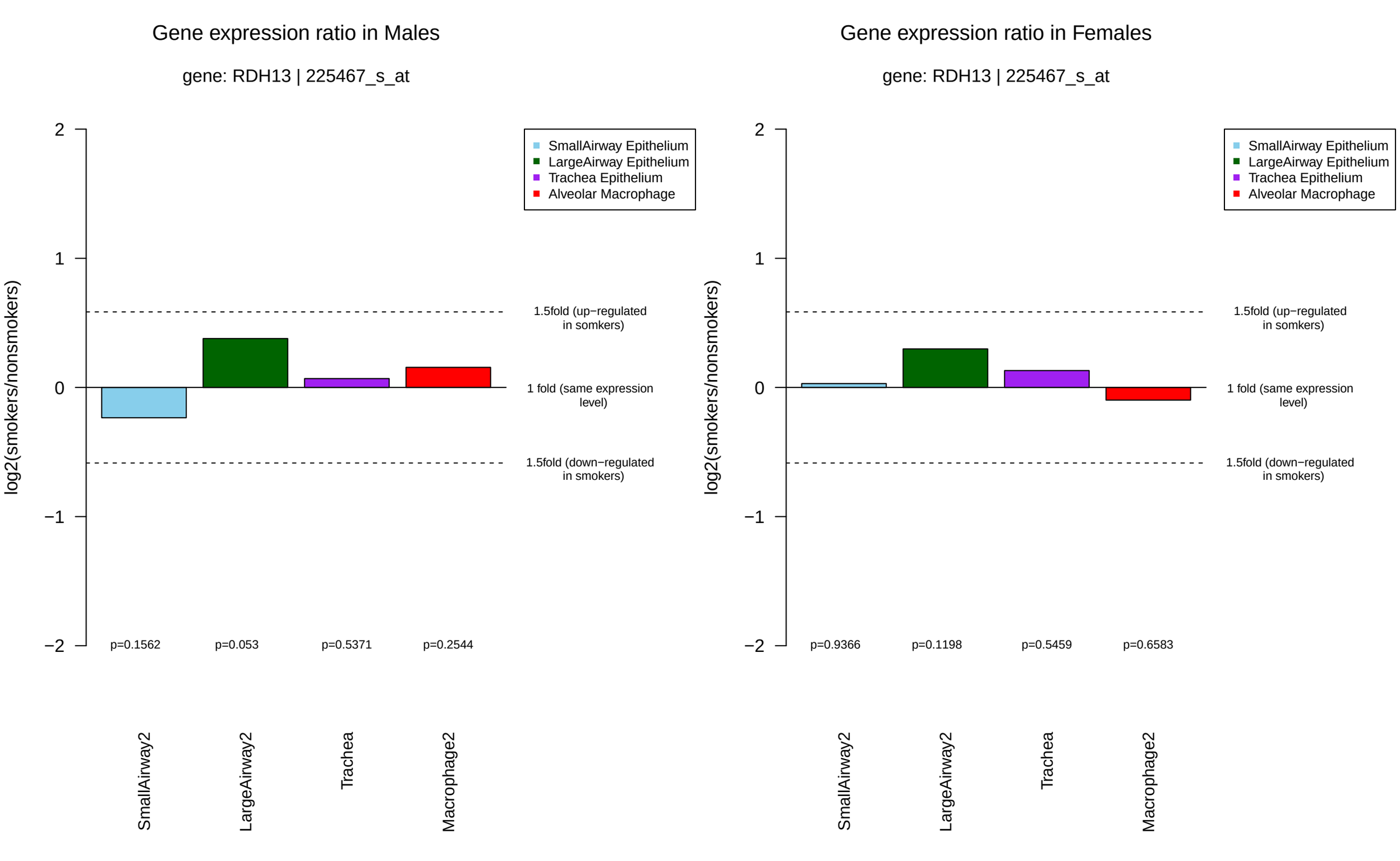
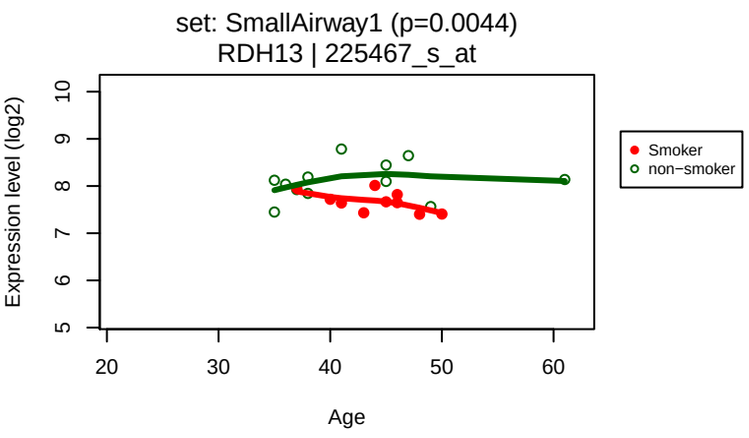
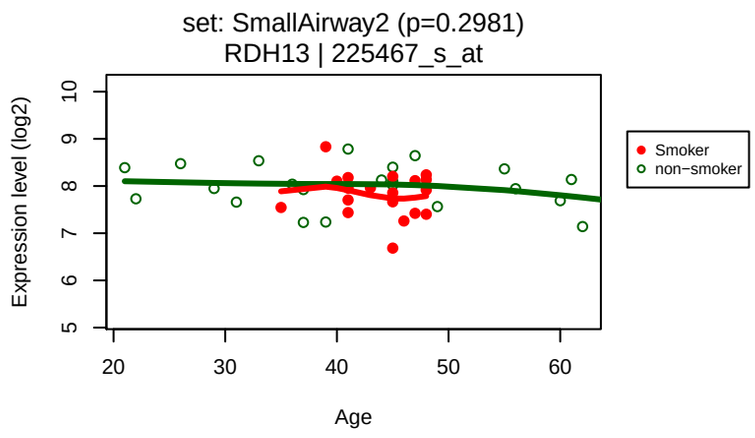
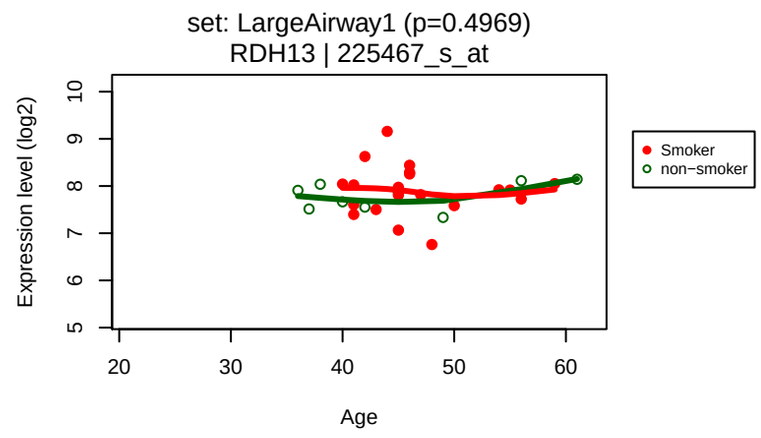

Gene expression of smokers and nonsmokers
gene: RDH13 | 225467_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0044
p=0.2981
p=0.7565
p=0.4969
p=0.0205
p=0.3497
p=0.2638
p=0.5269
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

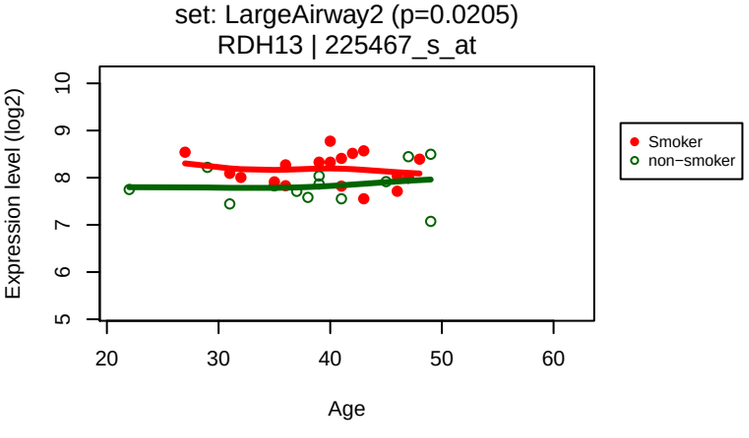

Gene expression ratio (smokers/nonsmokers)
gene: RDH13 | 225467_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0044
p=0.2981
p=0.7565
p=0.4969
p=0.0205
p=0.3497
p=0.2638
p=0.5269
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

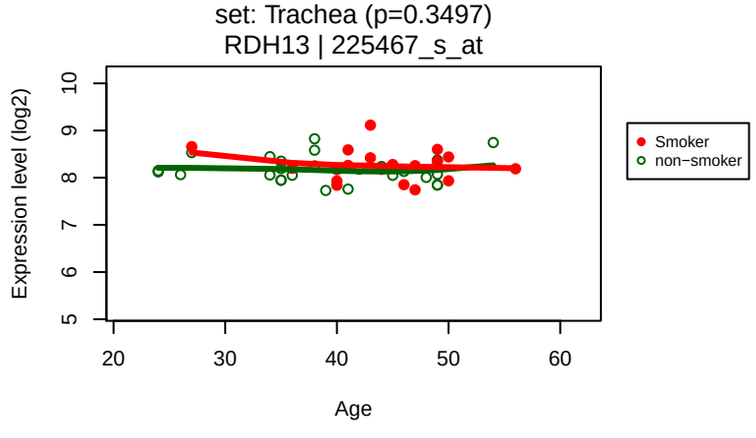

Gene expression ratio in Males
gene: RDH13 | 225467_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1562
p=0.053
p=0.5371
p=0.2544
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RDH13 | 225467_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9366
p=0.1198
p=0.5459
p=0.6583
SmallAirway2
LargeAirway2
Trachea
Macrophage2

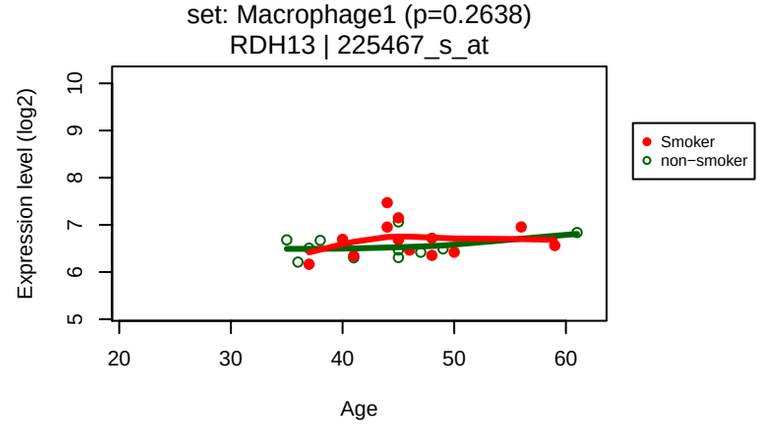

set: SmallAirway1 (p=0.0044)
RDH13 | 225467_s_at
Expression level (log2)
Age
Smoker
non−smoker

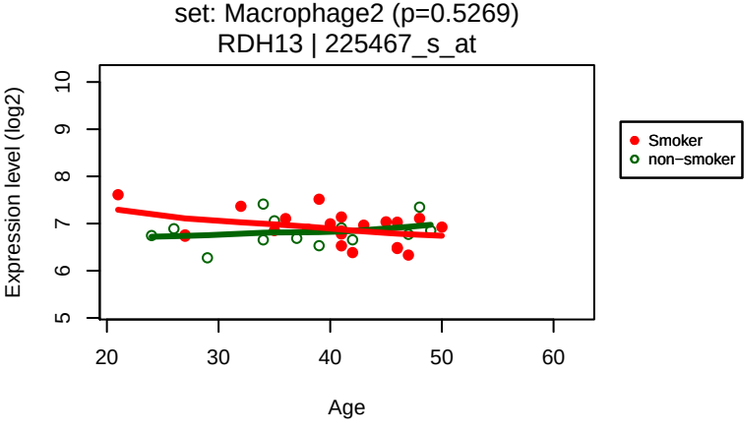

set: SmallAirway2 (p=0.2981)
RDH13 | 225467_s_at
Expression level (log2)
Age
Smoker
non−smoker

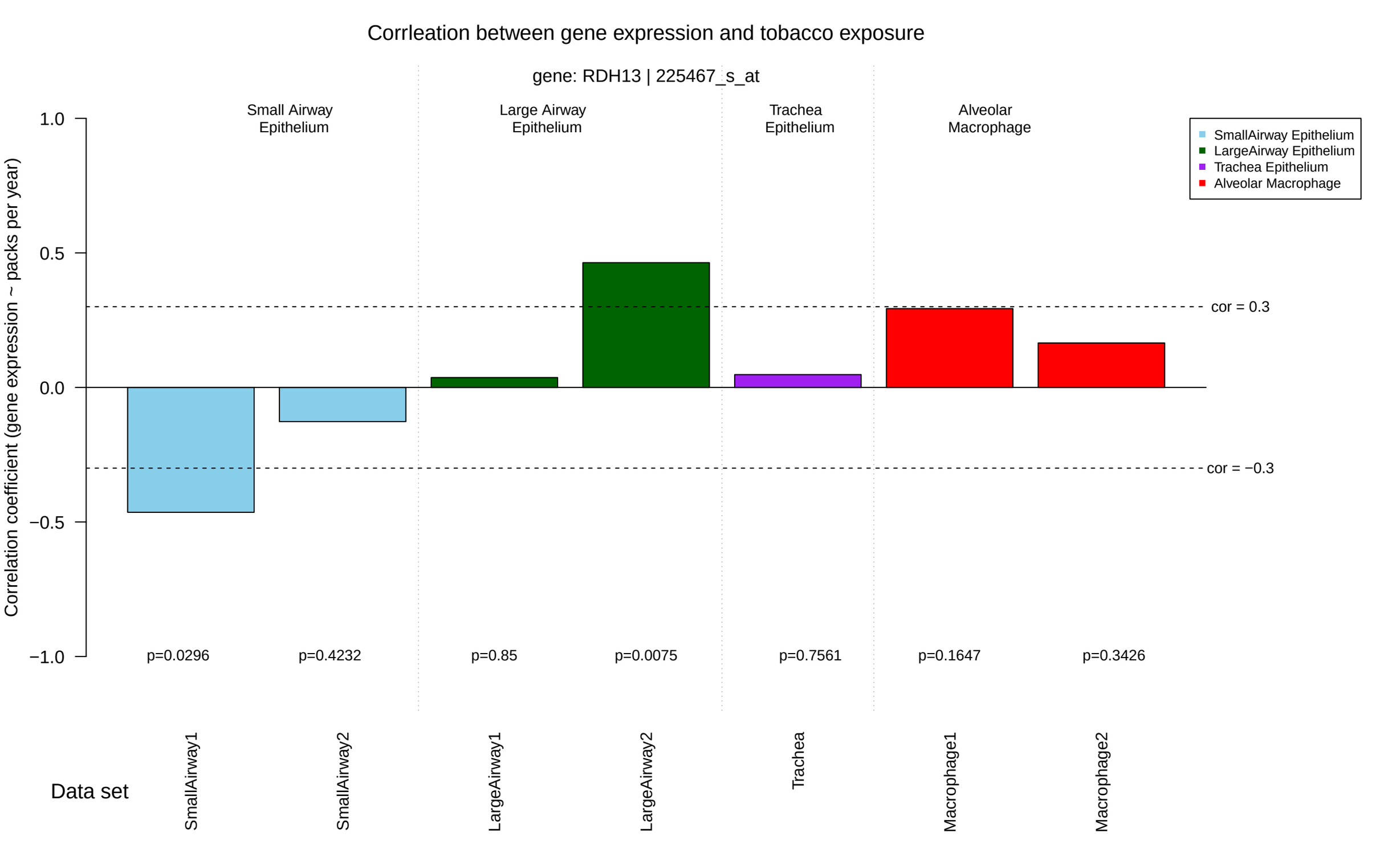

set: LargeAirway1 (p=0.4969)
RDH13 | 225467_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0205)
RDH13 | 225467_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3497)
RDH13 | 225467_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2638)
RDH13 | 225467_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5269)
RDH13 | 225467_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RDH13 | 225467_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0296
p=0.4232
p=0.85
p=0.0075
p=0.7561
p=0.1647
p=0.3426
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2